EC-360
SOFTWARE ENGINEERING

Lecture 8
User Interface Design

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Agenda

- Design principles
- GUI characteristics
- User Guidance
- UI Evaluation
- Examples

Design Principles

Principle	***Description***
User familiarity	Use terms and concepts *familiar* to the user.
Consistency	Comparable operations should be activated in the *same way*. Commands and menus should have the same format, etc.
Minimal surprise	If a command operates in a known way, the user should be able to *predict* the operation of comparable commands.
Feedback	Provide the user with visual and auditory feedback, maintaining *two-way communication*.

Design Principles

Principle	***Description***
Memory load	Reduce the amount of information that must be remembered between actions. *Minimize* the memory load.
Efficiency	Seek efficiency in dialogue, motion and thought. *Minimize keystrokes and mouse movements*.
Recoverability	Allow users to *recover from their errors*. Include undo facilities, confirmation of destructive actions, 'soft' deletes, etc.
User guidance	Incorporate some form of *context-sensitive user guidance* and assistance.

GUI Characteristics

Characteristic	***Description***
Windows	Multiple windows allow *different information to be displayed simultaneously* on the user's screen.
Icons	Usually icons represent *files* (including folders and applications), but they may also stand for *processes* (e.g., printer drivers).
Menus	Menus bundle and organize *commands* (eliminating the need for a command language).
Pointing	A pointing device such as a mouse is used for *command choices* from a menu or indicating items of interest in a window.
Graphics	Graphical elements can be *commands* on the same display.

GUI

Advantages

- They are *easy to learn* and use.
  - Users without experience can learn to use the system quickly.
- The user may *switch attention* between tasks and applications.
- *Fast, full-screen interaction* is possible with immediate access to the entire screen

Problems

- A GUI is not automatically a good interface
  - Many software systems are *never used* due to poor UI design
  - A poorly designed UI can cause a user to make *catastrophic errors*

Command Interfaces

Advantages

- Allow experienced users to *interact quickly* with the system

Problems

- Users have to *learn and remember* a command language
- Not suitable for *occasional* or inexperienced users
- An *error detection* and recovery system is required
- *Typing ability* is required

Direct Manipulation

Advantages

- Users *feel in control* and are less likely to be afraid by the system
- User *learning time* is relatively short
- Users get *immediate feedback* on their actions

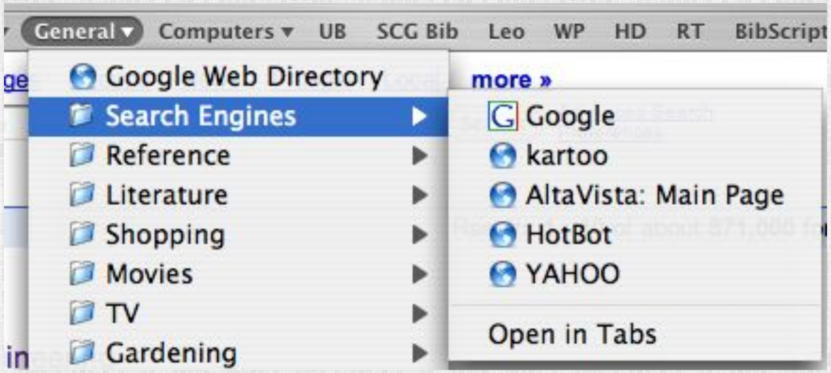
Problems

- Finding the right user *metaphor* may be difficult
- It can be *hard to navigate* efficiently in a large information space.
- It can be *complex to program* and demanding to execute

Menu Systems

Advantages

- Users don’t need to remember command names
- Typing effort is minimal
- User errors are trapped by the interface
- Context-dependent help can be provided (based on the current menu selection)



Problems

> If there are many choices, some *menu structuring* facility must be used

> *Experienced users find menus slower* than command language

Menu Structuring

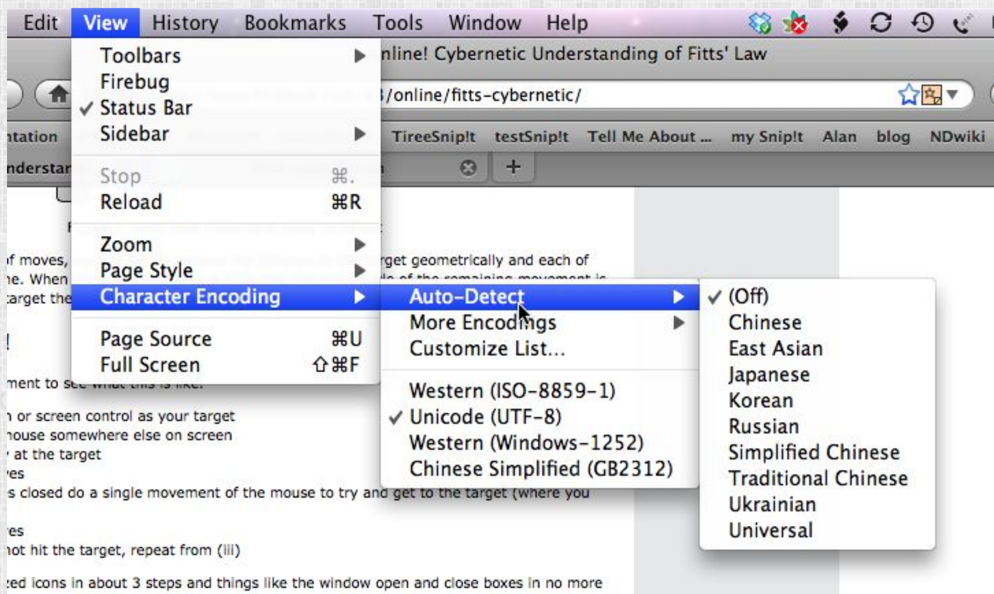
Scrolling menus

- The menu can be scrolled to reveal additional choices
- Not practical if there is a very large number of choices

Menu Structuring

Hierarchical menus

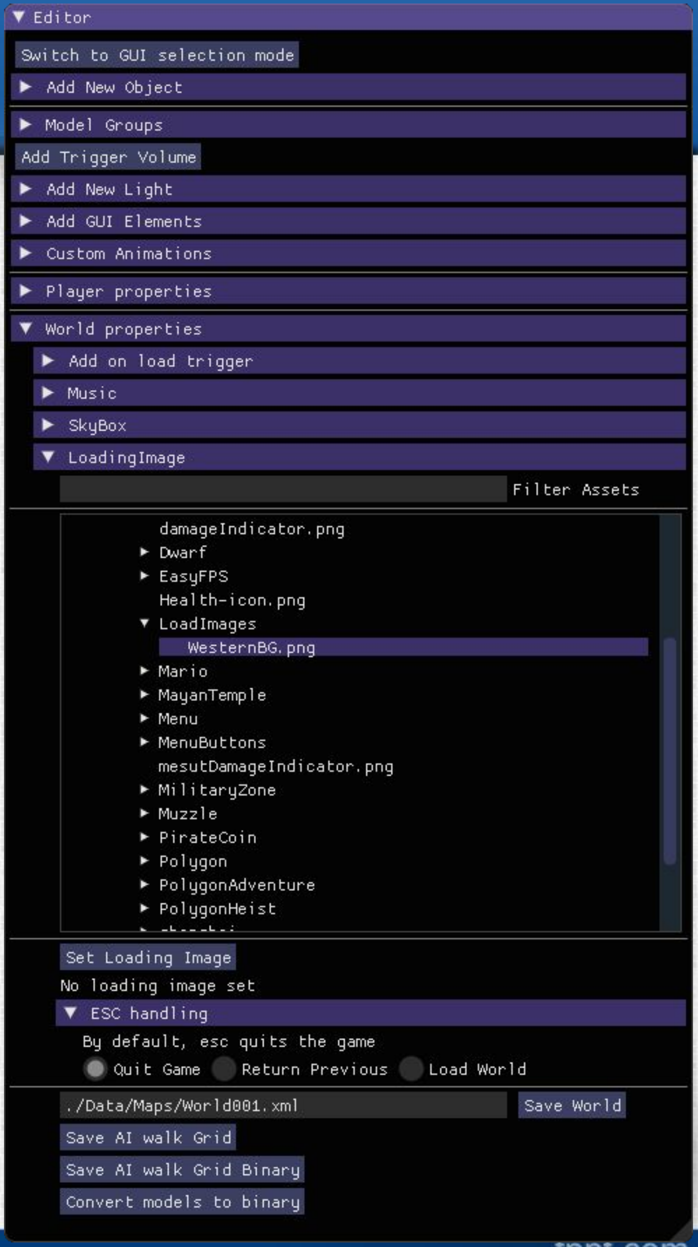
- Selecting a menu item causes the menu to be replaced by a sub-menu



Menu Structuring

Walking menus

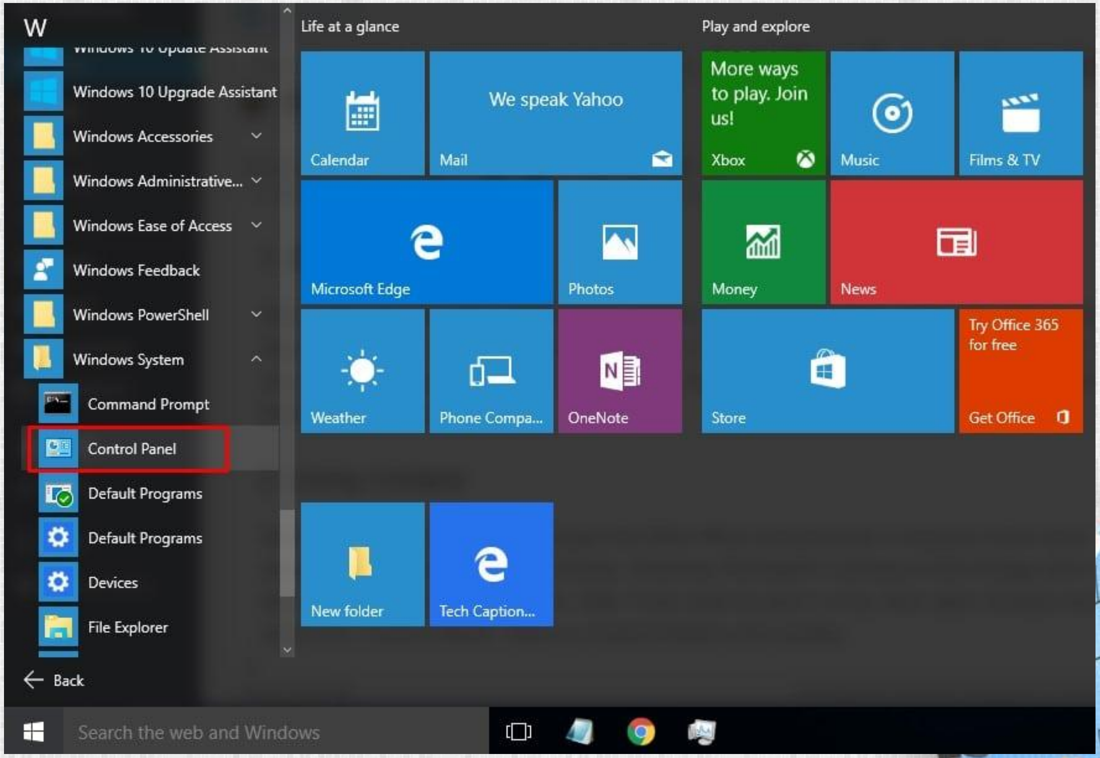
- A menu selection causes another menu to be revealed



Menu Structuring

Associated control panels

- When a menu item is selected, a control panel pops-up with further options



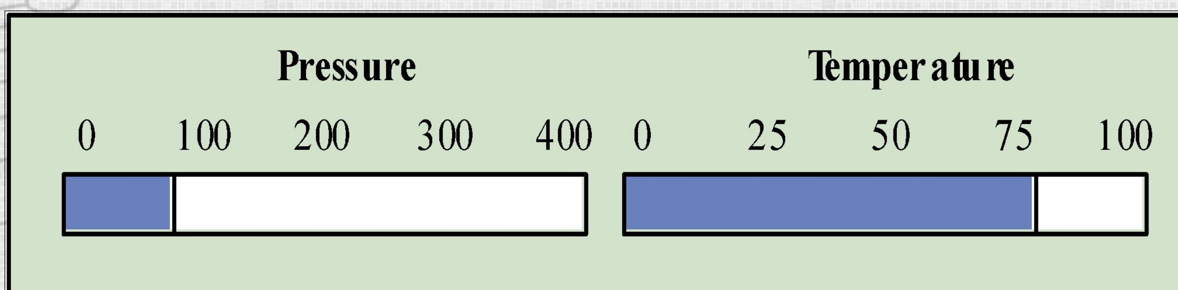
Analogue vs. Digital Presentation

Digital presentation

- *Compact* — takes up little screen space
- *Precise values* can be communicated

Analogue presentation

- Easier to get an 'at a glance' *impression* of a value
- Possible to show *relative values*
- Easier to see *exceptional* data values



Colour Use Guidelines

Colour can help the user *understand complex information structures.*

- Don’t use (only) colour to *communicate meaning*!
  - Open to *misinterpretation* (colour-blindness, cultural differences ...)
  - *Design for monochrome then add colour*
- Use colour coding to support user tasks
  - highlight exceptional events
  - allow users to control colour coding
- Use *colour change* to show *status change*
- Don't use too many colours
  - Avoid colour pairings which clash
- Use colour coding *consistently*

User Guidance

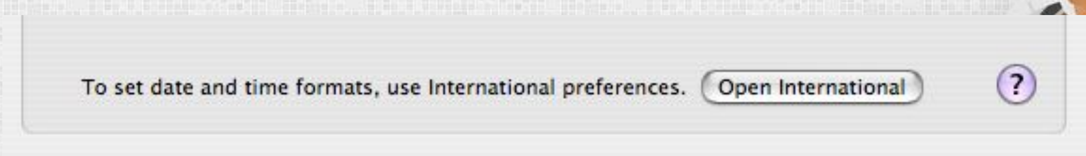
The <u>user guidance system</u> is integrated with the user interface to help users when they *need information* about the system or when they make some kind of *error*.

Includes

- System messages, including error messages
- Documentation provided for users
- On-line help

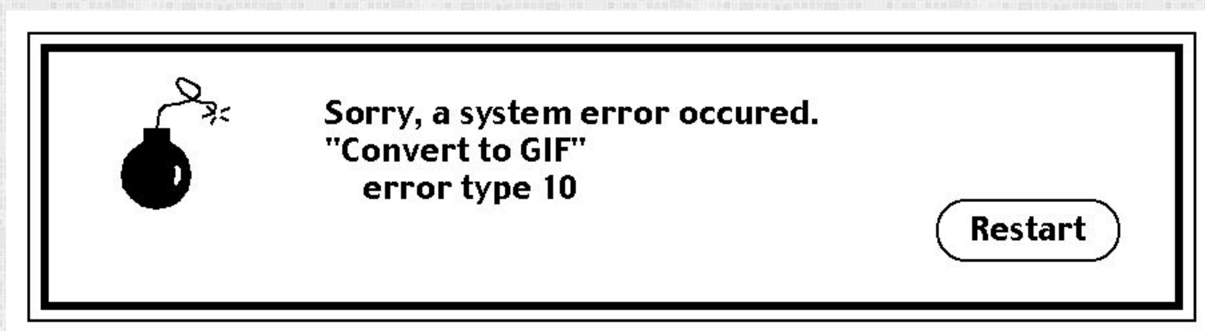
Help system use

- *Multiple entry points* should be provided
  - the user should be able to get help from different places
- The help system should indicate *where the user is positioned*
- *Navigation and traversal* facilities must be provided



Error Message Guidelines

- *Speak the user's language*
- Give *constructive advice* for recovering from the error
- Indicate *negative consequences* of the error (e.g., possibly corrupted files)
- Give an *audible or visual cue*
- Don't make the user feel guilty!

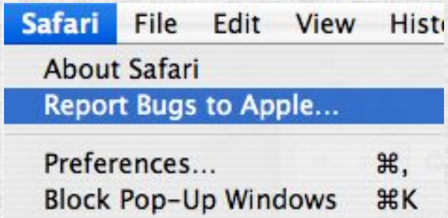


User interface evaluation

- Some evaluation of a user interface design should be carried out to assess its *usability*.
- Full scale evaluation is very *expensive* and *impractical* for most systems.
- Ideally, an interface should be evaluated against a *usability specification*. However, it is rare for such specifications to be produced.

Simple evaluation techniques

- *Questionnaires* for user feedback.
- *Video recording* of system use and subsequent tape evaluation.

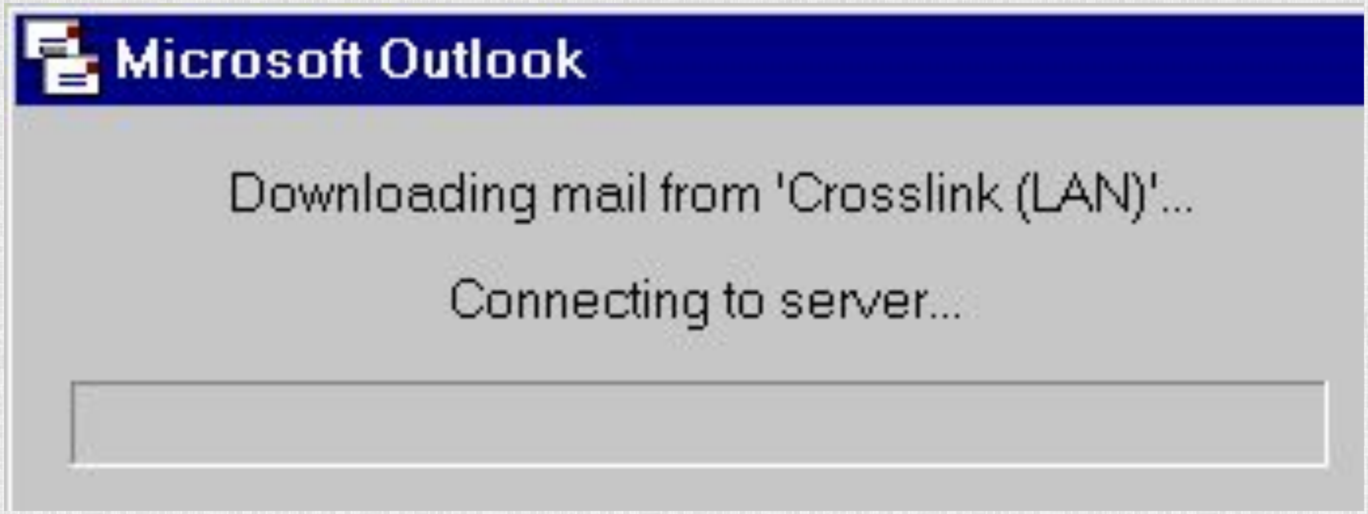


- *Instrumentation* of code to collect information about facility use and user errors.
- The provision of code in the software to collect *on-line user feedback*.

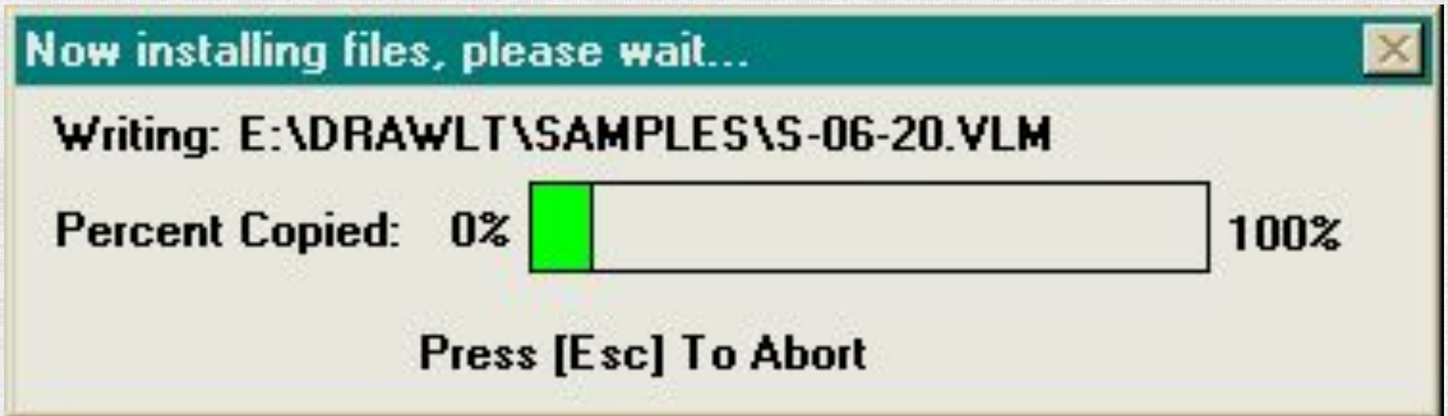
Usability Attributes

Attribute	***Description***
Learnability	How long does it take a new user to become *productive* with the system?
Speed of operation	How well does the system *response* match the user's work *practice*?
Robustness	How *tolerant* is the system of user error?
Recoverability	How good is the system at *recovering* from user errors?
Adaptability	How closely is the system tied to a *single model* of work?

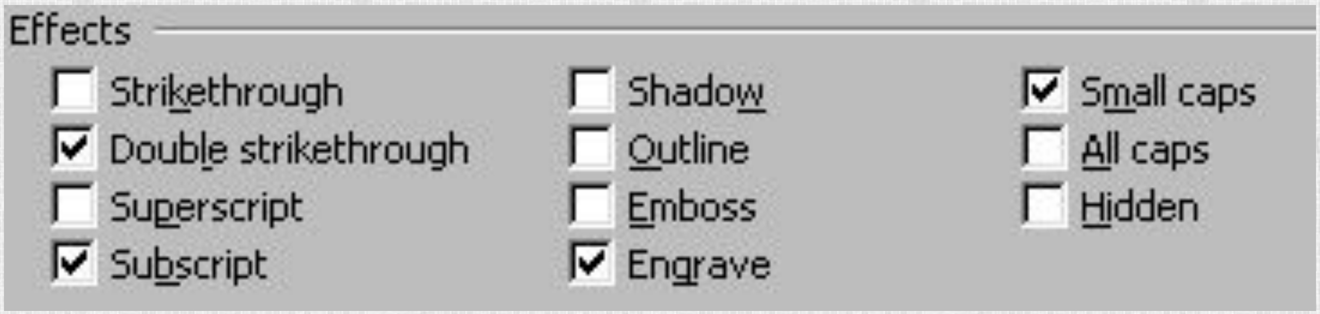
Is there progress?



Now, that's progress!



I want them all!

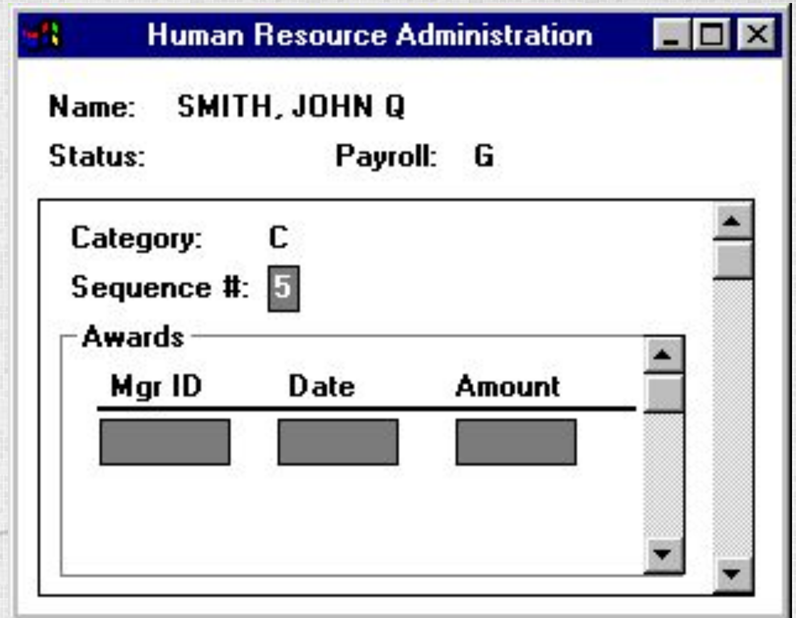


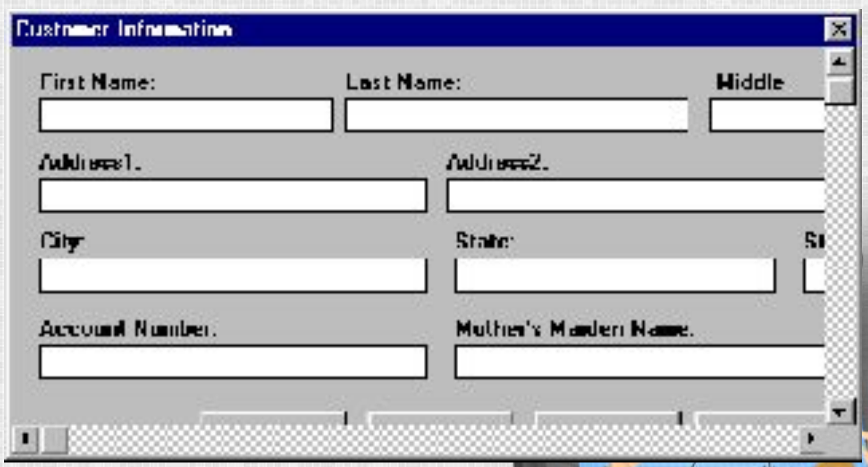
In Excel, "cut" doesn't mean cut

Region	January	February
North	10111	13400
South	22100	24050
East	13270	15670
West	10800	21500

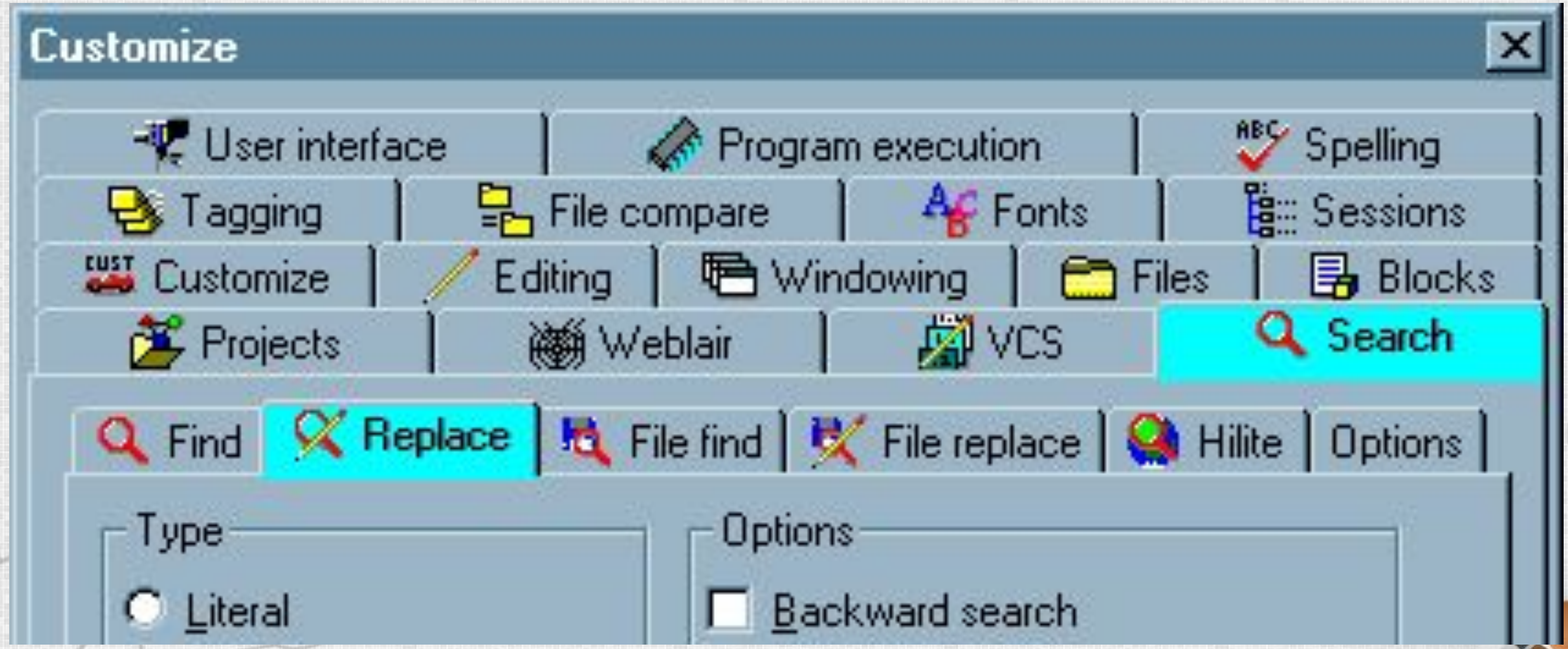
ESE 8.25

Fun with scrolling!

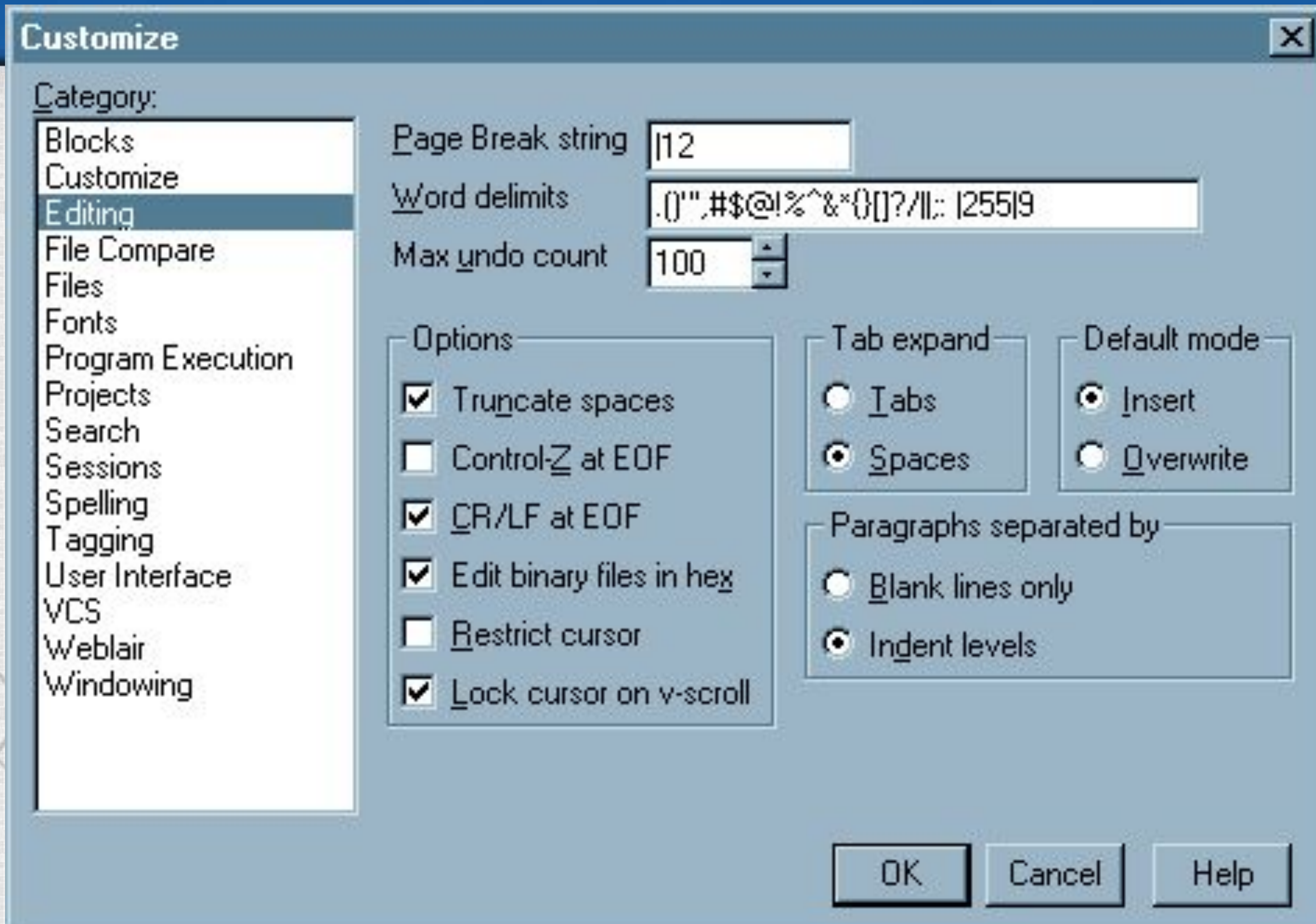




More tabs please!

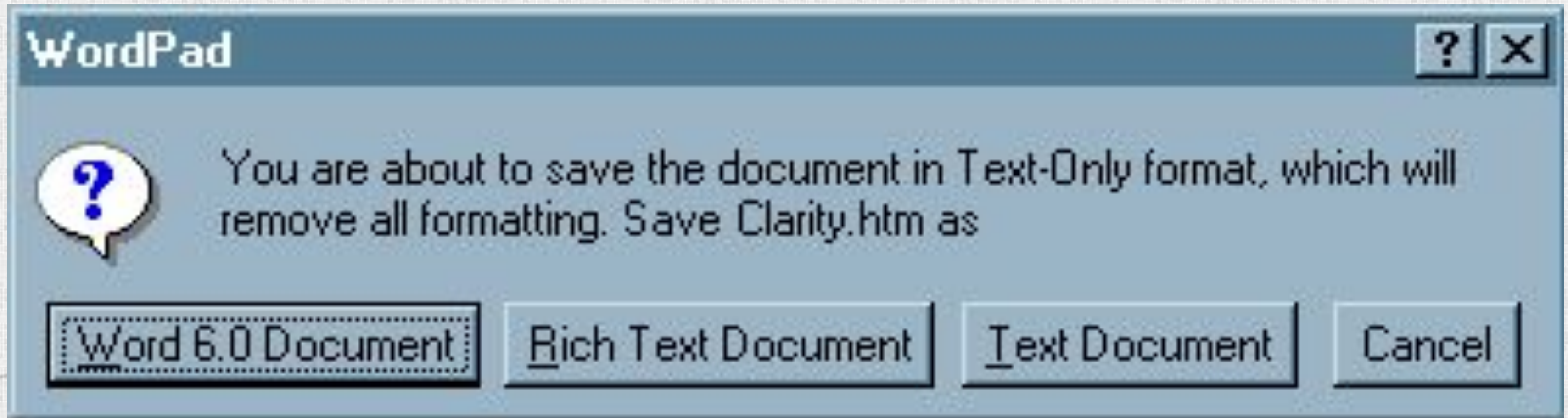


Without tabs



Helpful tips

I can’t make up my mind

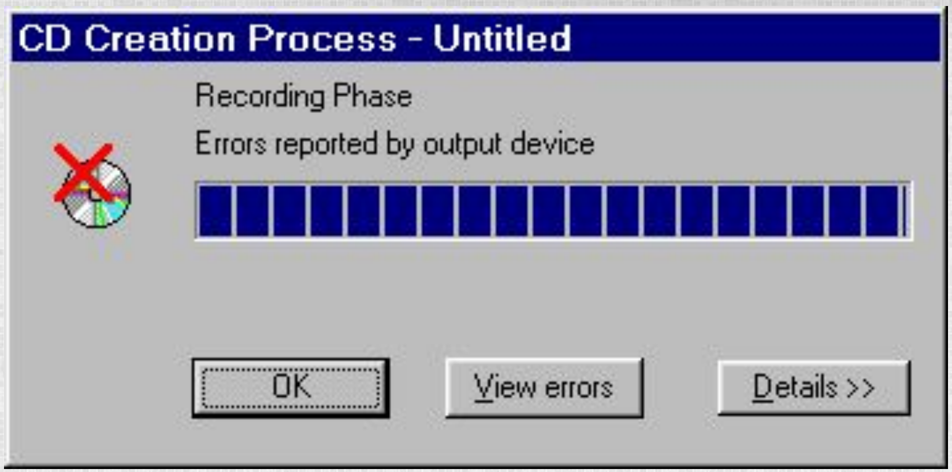


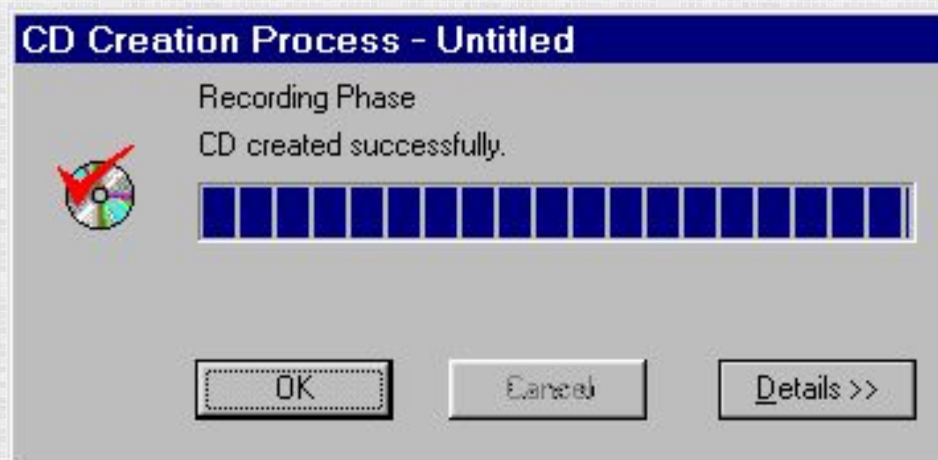
Whenever your local SMS Administrator sends you an actual software Package, the SMS Package Command Manager will appear (usually at network logon time) displaying the available Package(s). The following screenshots display scenes similar to what you will see when you receive an actual SMS Package.

To start the demonstration, click the "CLICK HERE" [illegible] bottom of the screen.

Green good — red bad

Was that an error?





Uh … ok

Yes — I mean, no

Key points

- *User interface design principles* should help guide the design of user interfaces.
- *Interaction styles* include direct manipulation, menu systems form fill-in, command languages and natural language.
- *Graphical displays* should be used to present trends and approximate values. *Digital displays* when precision is required.
- *Colour* should be used *sparingly and consistently*.
- The goals of *UI evaluation* are to *obtain feedback* on how to improve the interface design and to assess if the interface meets its *usability requirements*.

References

- *Software Engineering*, I. Sommerville,
- *Software Engineering — A Practitioner's Approach*, R. Pressman, Mc-Graw Hill